Chapter 1. Introduction

RP2350 is a new family of microcontrollers from Raspberry Pi that offers major enhancements over RP2040. Key features include:

- Dual Cortex-M33 or Hazard3 processors at 150 MHz
- 520 kB on-chip SRAM, in 10 independent banks
- 8 kB of one-time-programmable storage (OTP)
- Up to 16 MB of external QSPI flash or PSRAM through dedicated QSPI bus
- Additional 16 MB flash or PSRAM through optional second chip-select
- On-chip switched-mode power supply to generate core voltage
- Optional low-quiescent-current LDO mode for sleep states
- 2 × on-chip PLLs for internal or external clock generation
- GPIOs are 5 V-tolerant (powered) and 3.3 V-failsafe (unpowered)
- Security features:
  - Optional boot signing, enforced by on-chip mask ROM, with key fingerprint in OTP
  - Protected OTP storage for optional boot decryption key
  - Global bus filtering based on Arm or RISC-V security/privilege levels
  - Peripherals, GPIOs, and DMA channels individually assignable to security domains
  - Hardware mitigations for fault injection attacks
  - Hardware SHA-256 accelerator
- Peripherals:
  - 2 × UARTs
  - 2 × SPI controllers
  - 2 × I2C controllers
  - 24 × PWM channels
  - USB 1.1 controller and PHY, with host and device support
  - 12 × PIO state machines
  - 1 × HSTX peripheral

Table 1 shows the RP2350 family of devices, including options for QFN-80 (10 × 10 mm) and QFN-60 (7 × 7 mm) packages, with and without flash-in-package.

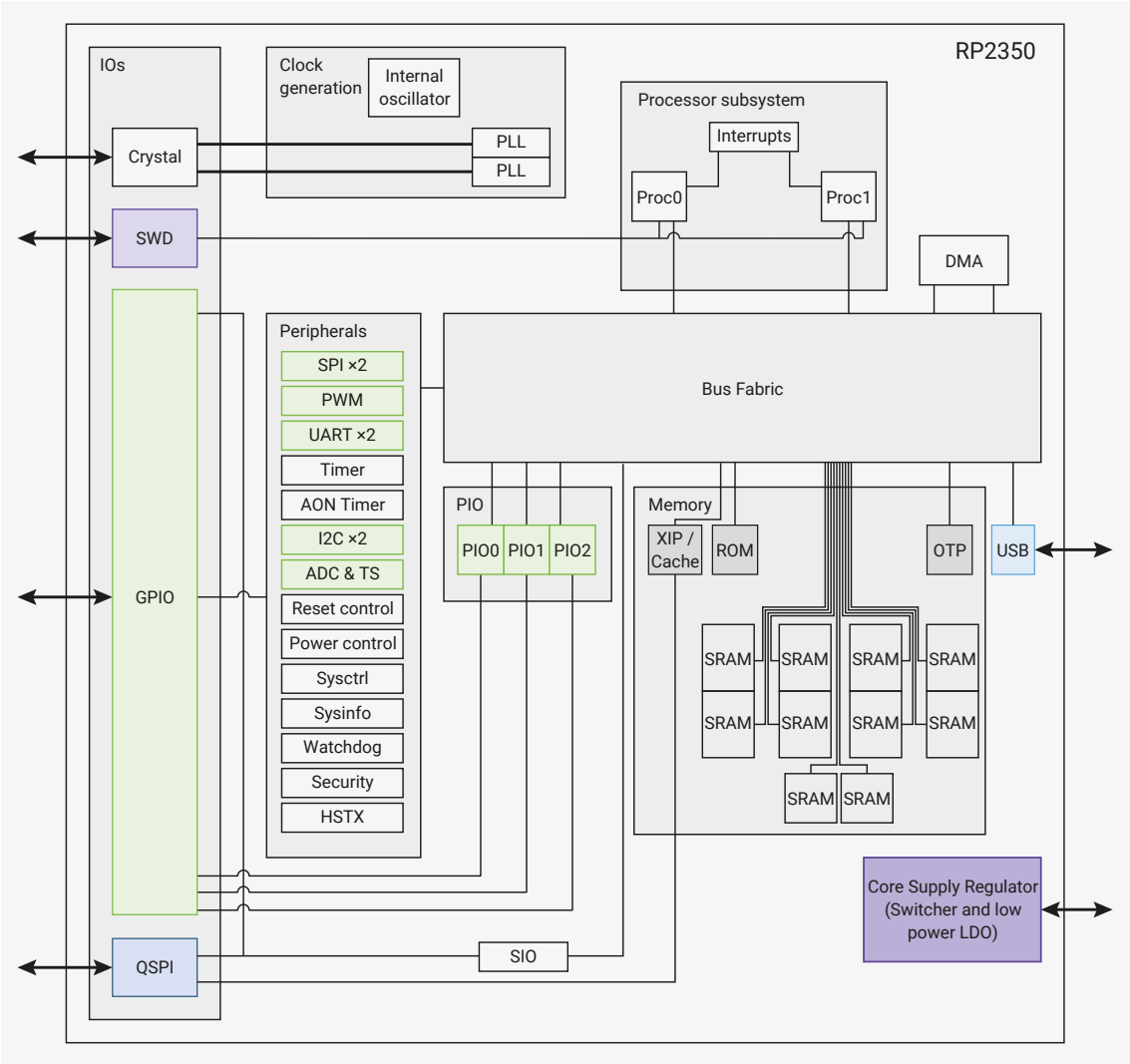
Table 1. RP2350 device family

Product	Package	Internal Flash	GPIO	Analogue Inputs
RP2350A	QFN-60	None	30	4
RP2350B	QFN-80	None	48	8
RP2354A	QFN-60	2 MB	30	4
RP2354B	QFN-80	2 MB	48	8

1.1. The chip

Dual Cortex-M33 or Hazard3 processors access RP2350's memory and peripherals via AHB and APB bus fabric.

Figure 1. A system overview of the RP2350 chip



Code may execute directly from external memory through a dedicated QSPI memory interface in the execute-in-place subsystem (XIP). The cache improves XIP performance significantly. Both flash and RAM can attach via this interface.

Debug is available via the SWD interface. This allows an external host to load, run, halt and inspect software running on the system, or configure the execution trace output.

Internal SRAM can contain code or data. It is addressed as a single 520 kB region, but physically partitioned into 10 banks to allow simultaneous parallel access from different managers. All SRAM supports single-cycle access.

A high-bandwidth system DMA offloads repetitive data transfer tasks from the processors.

GPIO pins can be driven directly via single-cycle IO (SIO), or from a variety of dedicated logic functions such as the hardware SPI, I2C, UART and PWM. Programmable IO controllers (PIO) can provide a wider variety of IO functions, or supplement the number of fixed-function peripherals.

A USB controller with embedded PHY provides FS/LS Host or Device connectivity under software control.

Four or eight ADC inputs (depending on package size) are shared with GPIO pins.

Two PLLs provide a fixed 48 MHz clock for USB or ADC, and a flexible system clock up to 150 MHz. A crystal oscillator provides a precise reference for the PLLs.

An internal voltage regulator supplies the core voltage, so you need generally only supply the IO voltage. It operates as a

switched mode buck converter when the system is awake, providing up to 200 mA at a variable output voltage, and can switch to a low-quiescent-current LDO mode when the system is asleep, providing up to 1 mA for state retention.

The system features low-power states where unused logic is powered off, supporting wakeup from timer or IO events. The amount of SRAM retained during power-down is configurable.

The internal 8 kB one-time-programmable storage (OTP) contains chip information such as unique identifiers, can be used to configure hardware and bootrom security features, and can be programmed with user-supplied code and data.

The built-in bootrom implements direct boot from flash or OTP, and serial boot from USB or UART. Code signature enforcement is supported for all boot media, using a key fingerprint registered in internal OTP storage. OTP can also store decryption keys for encrypted boot, preventing flash contents from being read externally.

RISC-V architecture support is implemented by dynamically swapping the Cortex-M33 (Armv8-M) processors with Hazard3 (RV32IMAC+) processors. Both architectures are available on all RP2350-family devices. The RISC-V cores support debug over SWD, and can be programmed with the same SDK as the Arm cores.

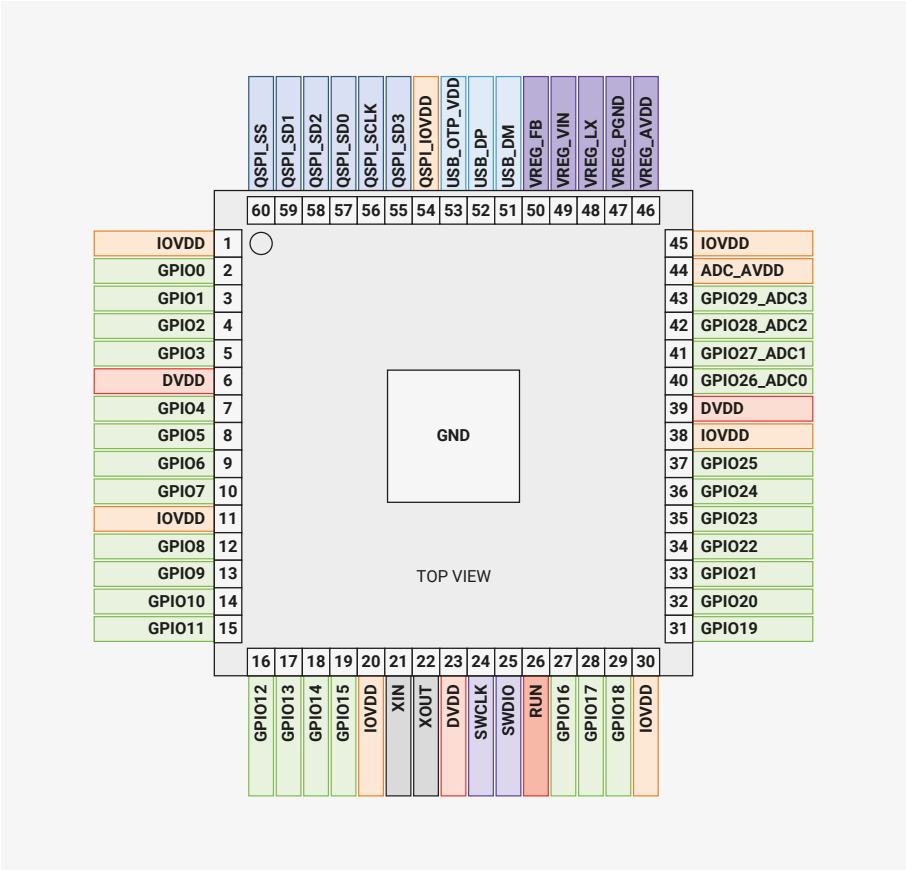
1.2. Pinout reference

This section provides a quick reference for pinout and pin functions. Full details, including electrical specifications and package drawings, can be found in Chapter 14.

1.2.1. Pin locations

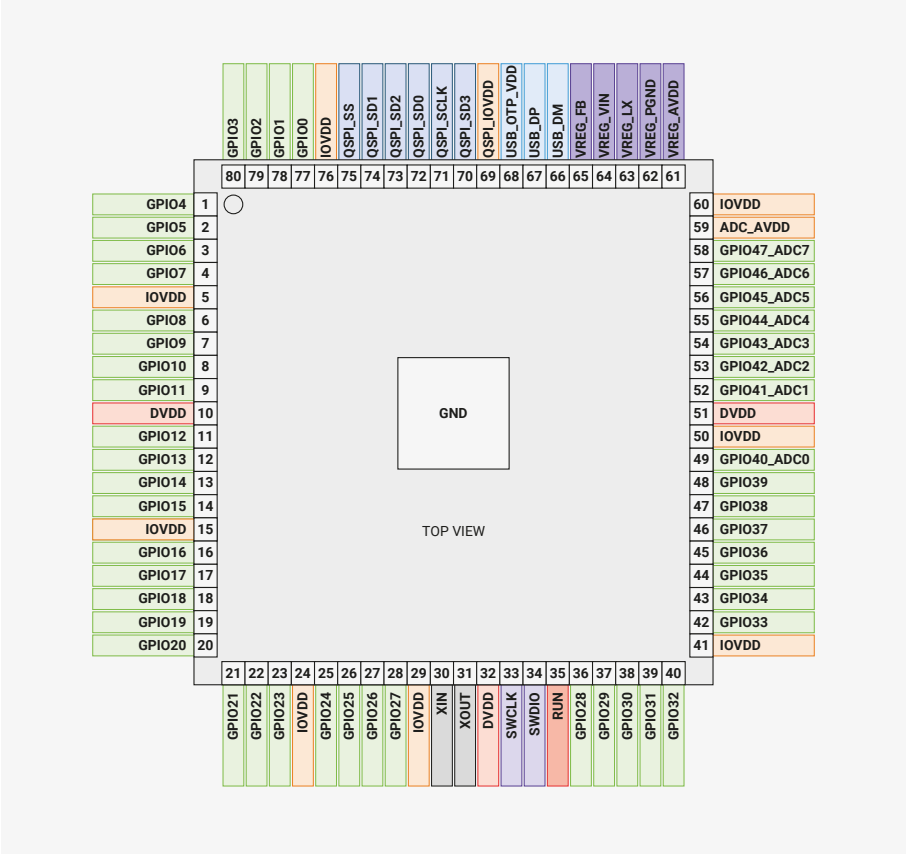
1.2.1.1. QFN-60 (RP2350A)

Figure 2. RP2350 Pinout for QFN-60 7×7mm (reduced ePad size)



1.2.1.2. QFN-80 (RP2350B)

Figure 3. RP2350 Pinout for QFN-80 10×10mm (reduced ePad size)



1.2.2. Pin descriptions

Table 2. The function of each pin is briefly described here. Full electrical specifications can be found in Chapter 14.

Name	Description
GPIOx	General-purpose digital input and output. RP2350 can connect one of a number of internal peripherals to each GPIO, or control GPIOs directly from software.
GPIOx/ADCy	General-purpose digital input and output, with analogue-to-digital converter function. The RP2350 ADC has an analogue multiplexer which can select any one of these pins, and sample the voltage.
QSPIx	Interface to a SPI, Dual-SPI or Quad-SPI flash or PSRAM device, with execute-in-place support. These pins can also be used as software-controlled GPIOs, if they are not required for flash access.
USB_DM and USB_DP	USB controller, supporting Full Speed device and Full/Low Speed host. A 27Ω series termination resistor is required on each pin, but bus pullups and pulldowns are provided internally. These pins can be used as software-controlled GPIOs, if USB is not required.
XIN and XOUT	Connect a crystal to RP2350's crystal oscillator. XIN can also be used as a single-ended CMOS clock input, with XOUT disconnected. The USB bootloader defaults to a 12MHz crystal or 12MHz clock input, but this can be configured via OTP.
RUN	Global asynchronous reset pin. Reset when driven low, run when driven high. If no external reset is required, this pin can be tied directly to IOVDD.
SWCLK and SWDIO	Access to the internal Serial Wire Debug multi-drop bus. Provides debug access to both processors, and can be used to download code.
GND	Single external ground connection, bonded to a number of internal ground pads on the RP2350 die.

Name	Description
IOVDD	Power supply for digital GPIOs, nominal voltage 1.8V to 3.3V
USB_OTP_VDD	Power supply for internal USB Full Speed PHY and OTP storage, nominal voltage 3.3V
ADC_AVDD	Power supply for analogue-to-digital converter, nominal voltage 3.3V
QSPI_IOVDD	Power supply for QSPI IOs, nominal voltage 1.8V to 3.3V
VREG_AVDD	Analogue power supply for internal core voltage regulator, nominal voltage 3.3V
VREG_PGND	Power-ground connection for internal core voltage regulator, tie to ground externally
VREG_LX	Switched-mode output for internal core voltage regulator, connected to external inductor. Max current 200 mA, nominal voltage 1.1V after filtering.
VREG_VIN	Power input for internal core voltage regulator, nominal voltage 2.7V to 5.5V
VREG_FB	Voltage feedback for internal core voltage regulator, connect to filtered VREG output (e.g. to DVDD, if the regulator is used to supply DVDD)
DVDD	Digital core power supply, nominal voltage 1.1V. Must be connected externally, either to the voltage regulator output, or an external board-level power supply.

1.2.3. GPIO functions (Bank 0)

Each individual GPIO pin can be connected to an internal peripheral via the GPIO functions defined below. Some internal peripheral connections appear in multiple places to allow some system level flexibility. SIO, PIO0, PIO1 and PIO2 can connect to all GPIO pins and are controlled by software (or software controlled state machines) so can be used to implement many functions.

Table 3. General Purpose Input/Output (GPIO) Bank 0 Functions

GPIO	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
0		SPI0 RX	UART0 TX	I2C0 SDA	PWM0 A	SIO	PIO0	PIO1	PIO2	QMI CS1n	USB OVCUR DET	
1		SPI0 CSn	UART0 RX	I2C0 SCL	PWM0 B	SIO	PIO0	PIO1	PIO2	TRACECLK	USB VBUS DET	
2		SPI0 SCK	UART0 CTS	I2C1 SDA	PWM1 A	SIO	PIO0	PIO1	PIO2	TRACEDATA0	USB VBUS EN	UART0 TX
3		SPI0 TX	UART0 RTS	I2C1 SCL	PWM1 B	SIO	PIO0	PIO1	PIO2	TRACEDATA1	USB OVCUR DET	UART0 RX
4		SPI0 RX	UART1 TX	I2C0 SDA	PWM2 A	SIO	PIO0	PIO1	PIO2	TRACEDATA2	USB VBUS DET	
5		SPI0 CSn	UART1 RX	I2C0 SCL	PWM2 B	SIO	PIO0	PIO1	PIO2	TRACEDATA3	USB VBUS EN	
6		SPI0 SCK	UART1 CTS	I2C1 SDA	PWM3 A	SIO	PIO0	PIO1	PIO2		USB OVCUR DET	UART1 TX
7		SPI0 TX	UART1 RTS	I2C1 SCL	PWM3 B	SIO	PIO0	PIO1	PIO2		USB VBUS DET	UART1 RX
8		SPI1 RX	UART1 TX	I2C0 SDA	PWM4 A	SIO	PIO0	PIO1	PIO2	QMI CS1n	USB VBUS EN	
9		SPI1 CSn	UART1 RX	I2C0 SCL	PWM4 B	SIO	PIO0	PIO1	PIO2		USB OVCUR DET	
10		SPI1 SCK	UART1 CTS	I2C1 SDA	PWM5 A	SIO	PIO0	PIO1	PIO2		USB VBUS DET	UART1 TX
11		SPI1 TX	UART1 RTS	I2C1 SCL	PWM5 B	SIO	PIO0	PIO1	PIO2		USB VBUS EN	UART1 RX
12	HSTX	SPI1 RX	UART0 TX	I2C0 SDA	PWM6 A	SIO	PIO0	PIO1	PIO2	CLOCK GPIN0	USB OVCUR DET	
13	HSTX	SPI1 CSn	UART0 RX	I2C0 SCL	PWM6 B	SIO	PIO0	PIO1	PIO2	CLOCK GPOUT0	USB VBUS DET	
14	HSTX	SPI1 SCK	UART0 CTS	I2C1 SDA	PWM7 A	SIO	PIO0	PIO1	PIO2	CLOCK GPIN1	USB VBUS EN	UART0 TX
15	HSTX	SPI1 TX	UART0 RTS	I2C1 SCL	PWM7 B	SIO	PIO0	PIO1	PIO2	CLOCK GPOUT1	USB OVCUR DET	UART0 RX
16	HSTX	SPI0 RX	UART0 TX	I2C0 SDA	PWM0 A	SIO	PIO0	PIO1	PIO2		USB VBUS DET	
17	HSTX	SPI0 CSn	UART0 RX	I2C0 SCL	PWM0 B	SIO	PIO0	PIO1	PIO2		USB VBUS EN	
18	HSTX	SPI0 SCK	UART0 CTS	I2C1 SDA	PWM1 A	SIO	PIO0	PIO1	PIO2		USB OVCUR DET	UART0 TX
19	HSTX	SPI0 TX	UART0 RTS	I2C1 SCL	PWM1 B	SIO	PIO0	PIO1	PIO2	QMI CS1n	USB VBUS DET	UART0 RX
20		SPI0 RX	UART1 TX	I2C0 SDA	PWM2 A	SIO	PIO0	PIO1	PIO2	CLOCK GPIN0	USB VBUS EN	
21		SPI0 CSn	UART1 RX	I2C0 SCL	PWM2 B	SIO	PIO0	PIO1	PIO2	CLOCK GPOUT0	USB OVCUR DET	
22		SPI0 SCK	UART1 CTS	I2C1 SDA	PWM3 A	SIO	PIO0	PIO1	PIO2	CLOCK GPIN1	USB VBUS DET	UART1 TX

GPIO	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
23		SPI0 TX	UART1 RTS	I2C1 SCL	PWM3 B	SIO	PIO0	PIO1	PIO2	CLOCK GPOUT1	USB VBUS EN	UART1 RX
24		SPI1 RX	UART1 TX	I2C0 SDA	PWM4 A	SIO	PIO0	PIO1	PIO2	CLOCK GPOUT2	USB OVCUR DET	
25		SPI1 CSn	UART1 RX	I2C0 SCL	PWM4 B	SIO	PIO0	PIO1	PIO2	CLOCK GPOUT3	USB VBUS DET	
26		SPI1 SCK	UART1 CTS	I2C1 SDA	PWM5 A	SIO	PIO0	PIO1	PIO2		USB VBUS EN	UART1 TX
27		SPI1 TX	UART1 RTS	I2C1 SCL	PWM5 B	SIO	PIO0	PIO1	PIO2		USB OVCUR DET	UART1 RX
28		SPI1 RX	UART0 TX	I2C0 SDA	PWM6 A	SIO	PIO0	PIO1	PIO2		USB VBUS DET	
29		SPI1 CSn	UART0 RX	I2C0 SCL	PWM6 B	SIO	PIO0	PIO1	PIO2		USB VBUS EN	
GPIOs 30 through 47 are QFN-80 only:												
30		SPI1 SCK	UART0 CTS	I2C1 SDA	PWM7 A	SIO	PIO0	PIO1	PIO2		USB OVCUR DET	UART0 TX
31		SPI1 TX	UART0 RTS	I2C1 SCL	PWM7 B	SIO	PIO0	PIO1	PIO2		USB VBUS DET	UART0 RX
32		SPI0 RX	UART0 TX	I2C0 SDA	PWM8 A	SIO	PIO0	PIO1	PIO2		USB VBUS EN	
33		SPI0 CSn	UART0 RX	I2C0 SCL	PWM8 B	SIO	PIO0	PIO1	PIO2		USB OVCUR DET	
34		SPI0 SCK	UART0 CTS	I2C1 SDA	PWM9 A	SIO	PIO0	PIO1	PIO2		USB VBUS DET	UART0 TX
35		SPI0 TX	UART0 RTS	I2C1 SCL	PWM9 B	SIO	PIO0	PIO1	PIO2		USB VBUS EN	UART0 RX
36		SPI0 RX	UART1 TX	I2C0 SDA	PWM10 A	SIO	PIO0	PIO1	PIO2		USB OVCUR DET	
37		SPI0 CSn	UART1 RX	I2C0 SCL	PWM10 B	SIO	PIO0	PIO1	PIO2		USB VBUS DET	
38		SPI0 SCK	UART1 CTS	I2C1 SDA	PWM11 A	SIO	PIO0	PIO1	PIO2		USB VBUS EN	UART1 TX
39		SPI0 TX	UART1 RTS	I2C1 SCL	PWM11 B	SIO	PIO0	PIO1	PIO2		USB OVCUR DET	UART1 RX
40		SPI1 RX	UART1 TX	I2C0 SDA	PWM8 A	SIO	PIO0	PIO1	PIO2		USB VBUS DET	
41		SPI1 CSn	UART1 RX	I2C0 SCL	PWM8 B	SIO	PIO0	PIO1	PIO2		USB VBUS EN	
42		SPI1 SCK	UART1 CTS	I2C1 SDA	PWM9 A	SIO	PIO0	PIO1	PIO2		USB OVCUR DET	UART1 TX
43		SPI1 TX	UART1 RTS	I2C1 SCL	PWM9 B	SIO	PIO0	PIO1	PIO2		USB VBUS DET	UART1 RX
44		SPI1 RX	UART0 TX	I2C0 SDA	PWM10 A	SIO	PIO0	PIO1	PIO2		USB VBUS EN	

GPIO	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
45		SPI1 CSn	UART0 RX	I2C0 SCL	PWM10 B	SIO	PIO0	PIO1	PIO2		USB OVCUR DET	
46		SPI1 SCK	UART0 CTS	I2C1 SDA	PWM11 A	SIO	PIO0	PIO1	PIO2		USB VBUS DET	UART0 TX
47		SPI1 TX	UART0 RTS	I2C1 SCL	PWM11 B	SIO	PIO0	PIO1	PIO2	QMI CS1n	USB VBUS EN	UART0 RX

Table 4. GPIO bank 0 function descriptions

Function Name	Description
SPIx	Connect one of the internal PL022 SPI peripherals to GPIO
UARTx	Connect one of the internal PL011 UART peripherals to GPIO
I2Cx	Connect one of the internal DW I2C peripherals to GPIO
PWMx A/B	Connect a PWM slice to GPIO. There are twelve PWM slices, each with two output channels (A/B). The B pin can also be used as an input, for frequency and duty cycle measurement.
SIO	Software control of GPIO, from the single-cycle IO (SIO) block. The SIO function (F5) must be selected for the processors to *drive* a GPIO, but the input is always connected, so software can check the state of GPIOs at any time.
PIOx	Connect one of the programmable IO blocks (PIO) to GPIO. PIO can implement a *wide* variety of interfaces, and has its own internal pin mapping hardware, allowing flexible placement of digital interfaces on bank 0 GPIOs. The PIO function (F6, F7, F8) must be selected for PIO to *drive* a GPIO, but the input is always connected, so the PIOs can always see the state of all pins.
HSTX	Connect the high-speed transmit peripheral (HSTX) to GPIO
CLOCK GPINx	General purpose clock inputs. Can be routed to a number of internal clock domains on RP2350, e.g. to provide a 1Hz clock for the AON Timer, or can be connected to an internal frequency counter.
CLOCK GPOUTx	General purpose clock outputs. Can drive a number of internal clocks (including PLL outputs) onto GPIOs, with optional integer divide.
TRACECLK, TRACEDATAx	CoreSight TPIU execution trace output from Cortex-M33 processors (Arm-only)
USB OVCUR DET/VBUS DET/VBUS EN	USB power control signals to/from the internal USB controller
QMI CS1n	Auxiliary chip select for QSPI bus, to allow execute-in-place from an additional flash or PSRAM device

NOTE

GPIOs 0 through 29 are available in all package variants. GPIOs 30 through 47 are available only in QFN-80 (RP2350B) package.

NOTE

Analogue input is available on GPIOs 26 through 29 in the QFN-60 package (RP2350A), for a total of four inputs, and on GPIOs 40 through 47 in the QFN-80 package (RP2350B), for a total of eight inputs.

1.2.4. GPIO functions (Bank 1)

GPIO functions are also available on the six dedicated QSPI pins, which are usually used for flash execute-in-place, and on the USB DP/DM pins. These may become available for general-purpose use depending on the use case, for example, QSPI pins may not be needed for code execution if RP2350 is booting from internal OTP storage, or being controlled externally via SWD.

Table 5. GPIO Bank 1 Functions

Pin	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
USB DP			UART1 TX	I2C0 SDA		SIO						

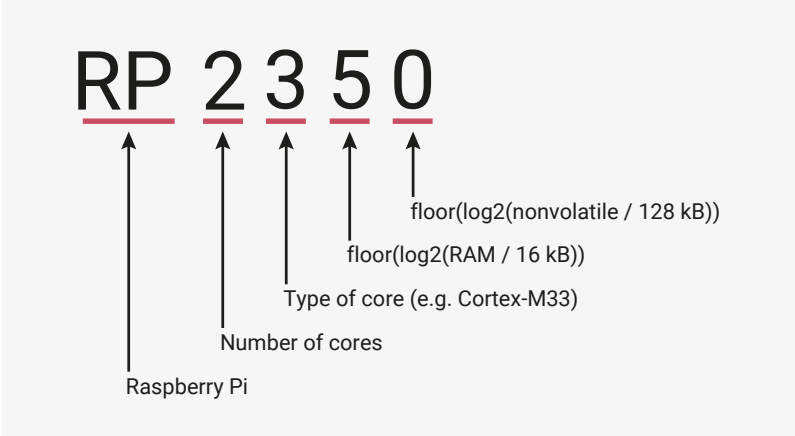
Pin	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
USB DM			UART1 RX	I2C0 SCL		SIO						
QSPI SCK	QMI SCK		UART1 CTS	I2C1 SDA		SIO						UART1 TX
QSPI CSn	QMI CS0n		UART1 RTS	I2C1 SCL		SIO						UART1 RX
QSPI SD0	QMI SD0		UART0 TX	I2C0 SDA		SIO						
QSPI SD1	QMI SD1		UART0 RX	I2C0 SCL		SIO						
QSPI SD2	QMI SD2		UART0 CTS	I2C1 SDA		SIO						UART0 TX
QSPI SD3	QMI SD3		UART0 RTS	I2C1 SCL		SIO						UART0 RX

Table 6. GPIO bank 1 function descriptions

Function Name	Description
UARTx	Connect one of the internal PL011 UART peripherals to GPIO
I2Cx	Connect one of the internal DW I2C peripherals to GPIO
SIO	Software control of GPIO, from the single-cycle IO (SIO) block. The SIO function (F5) must be selected for the processors to *drive* a GPIO, but the input is always connected, so software can check the state of GPIOs at any time.
QMI	QSPI memory interface peripheral, used for execute-in-place from external QSPI flash or PSRAM memory devices.

1.3. Why is the chip called RP2350?

Figure 4. An explanation for the name of the RP2350 chip.



The post-fix numeral on RP2350 comes from the following,

1. Number of processor cores
   - **2** indicates a dual-core system
2. Loosely which type of processor
   - **3** indicates Cortex-M33 or Hazard3
3. Internal memory capacity: $\log_2 \lfloor \frac{RAM}{16kB} \rfloor$
   - **5** indicates at least $2^5 \times 16$ kB = 512 kB
   - RP2350 has 520 kB of main system SRAM
4. Internal storage capacity: $\log_2 \lfloor \frac{nonvolatile}{128kB} \rfloor$ (or 0 if no onboard nonvolatile storage)

- RP235**0** uses external flash
- RP235**4** has $2^4 \times 128$ kB = 2 MB of internal flash

1.4. Version History

Table 7 lists versions of RP2350. Later versions fix bugs in earlier versions. For more information about the changes made between versions, see Appendix C. Also refer to Product Change Notification (PCN) 28.

Table 7. RP2350 version history

Version	Use
A0	Internal development
A1	Internal development
A2	Initial release
A3	Internal development, samples, and limited production
A4	Production version